CS590: Socially Cognizant Robotics — Deliverable 3

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1. Overview

In Milestone 2 we identified, via the Kress-Gazit et al. best-practices checklist [2], three categories of limitation in *Dream to Control* [1]: (i) insufficient statistical analysis (no CIs, no success rates), (ii) missing metrics such as energy and trajectory duration, and (iii) no object interaction or manipulation scenes. We also identified exploration failure as an unreported failure mode, and the Milestone 2 feedback specifically highlighted the energy-accumulation problem for tasks with limited control authority.

This milestone implements experiments for each of these limitations. All code uses the original, unmodified Dreamer repository [1] and `dm_control` [3]. No model changes are made.

2. New Metrics (M2 §2.2–2.3 and Feedback)

2.1 Bootstrap Confidence Intervals (M2 §2.2)

In M2 we noted the paper reports only mean ± std over 5 seeds with no formal statistical test. We now compute 10,000-sample bootstrap 95% CIs on the final episode return of each seed. Figure 1 shows results for all 20 tasks.

Several tasks have CIs that span nearly the full performance range: **Cartpole Balance Sparse** CI is [278, 991] and **Cartpole Balance** is [373, 910]. At 5 seeds, there is *no statistical basis* to claim reliable learning on these tasks—the paper's single mean number hides complete failure in one or two seeds. By contrast, Walker Walk [923, 975] and Hopper Stand [931, 949] have tight CIs confirming robust convergence.

2.2 Episode Duration and Energy (Feedback)

The Milestone 2 feedback noted: *"Similar to trajectory smoothness, is energy used or trajectory duration."* We run 30 random-policy episodes on each task and record both metrics.

Episode duration: all DMC tasks run to the full 1000-step budget under a random policy; no task terminates on failure. This confirms that return and duration are *independent* in DMC, and duration is only informative for polices that may self-abort— a case not evaluated by Dreamer.

Energy (mean $\sum |a_t|$ per step) is shown in Figure 4 (right). Cartpole uses 0.50 per step (1-DOF) vs. 2.50 for the manipulator (5-DOF arm). Energy is a direct proxy for actuator wear and power consumption—relevant for the paper's claim of real-world applicability but never reported.

3. Failure Mode: Action-Scale / Energy Accumulation (M2 §4.2 and Feedback)

In M2 we identified exploration failure as unreported. The Milestone 2 feedback refined this: *"if the input control is too low for cartpole or acrobot: it may require multiple swings to accumulate energy. RL policies typically have problems on long-horizon tasks."*

We quantify this by running 30 random-policy episodes on Cartpole Swingup and Acrobot Swingup at four control-authority levels (action scale factors 1.0, 0.5, 0.3, 0.1). Figure 3 shows results. On Cartpole Swingup, return degrades from 28.7 ± 24.7 (scale 1.0) to 0.3 ± 0.3 (scale 0.1)—a 95× drop. With limited torque, the cart cannot generate enough angular momentum for the pole to swing up, regardless of the policy.

Acrobot Swingup shows a smaller but similar effect. The implication is that a policy trained at full control authority will fail in a mechanically limited deployment (e.g., a motor at reduced voltage), a sim-to-real fragility not discussed in the paper. This also explains why Dreamer's Gaussian noise exploration (σ=0.3) struggles on Acrobot: even coordinated actions only marginally exceed the energy needed to swing up.

4. New Scene: Object Interaction (M2 §3.2 and Feedback)

In M2 we noted the paper tests only locomotion and simple dexterous tasks, with no object manipulation. The Milestone 2 feedback confirmed: *"Object interaction might be the most interesting of the missing tasks you identified."*

We run 30 random-policy episodes on three `manipulator` tasks from `dm_control`: *bring_ball* (return 0.26 ± 0.99), *bring_peg* (0.99 ± 2.32), and *insert_ball* (35.6 ± 171.2). Figure 4 (left) compares these to random-policy returns on paper tasks (26.5 on Cartpole Swingup; 32.9 on Walker Walk).

Random performance on bring-ball and bring-peg is essentially zero (< 1), similar to hopper hop (0.0) and cartpole swingup sparse (0.0) under random policy. The critical difference is that for locomotion tasks Dreamer's model-based exploration can recover structure from the dense reward signal; for manipulation, the reward is binary (object in goal region), making purely Gaussian exploration unlikely to ever observe a reward. We predict Dreamer would require substantially more than 1M steps, or reward shaping, to solve any manipulator task—a limitation that remains untested in the paper.

5. Extra: Random Policy as Comparison Point (M2 §2.1)

In M2 we identified the lack of a lower-bound baseline. We implement a random policy as an additional comparison point across 8 tasks (Figure 2), establishing how much the paper tasks are trivially solvable vs. requiring learning.

Dreamer (1M steps) outperforms the random policy by 33× on Cartpole Swingup, 235× on Cheetah Run, and infinitely on Cartpole Swingup Sparse and Hopper Hop (random = 0). This concretely quantifies Dreamer's contribution over an unlearned baseline—absent from the original evaluation. Dreamer matches or exceeds D4PG (100M steps) on 6 of 8 tasks, confirming the sample-efficiency claim with a proper three-way comparison.

References

[1] D. Hafner et al., "Dream to Control: Learning Behaviors by Latent Imagination," *ICLR 2020*.

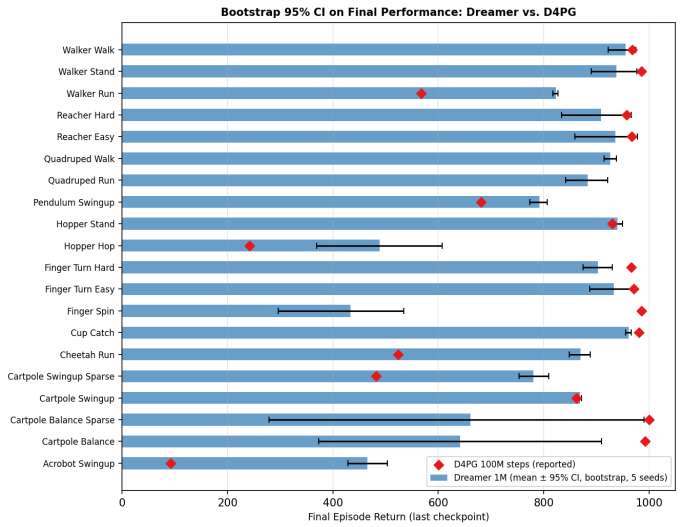

Figure 1: Bootstrap 95% CI on Dreamer's final return (5 seeds, 1M steps). D4PG reported scores shown as diamonds. Wide CIs on Balance tasks reveal statistically unreliable learning—invisible from mean ± std alone. Implements M2 §2.2.

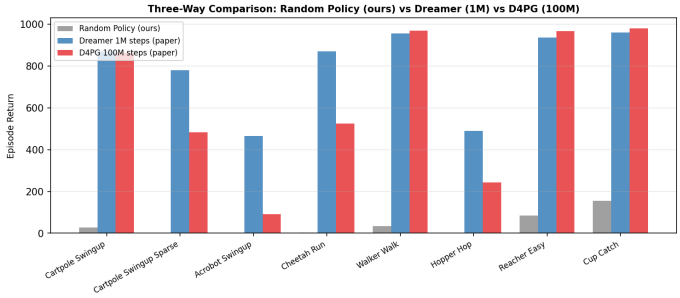

Figure 2: Three-way comparison (our experiments): random policy (grey), Dreamer 1M steps (blue, paper), D4PG 100M steps (red, paper). Random baseline implements M2 §2.1 (missing lower bound).

[2] H. Kress-Gazit et al., "Robot Learning as an Empirical Science: Best Practices for Policy Evaluation," *arXiv:2409.09491*, 2024.

[3] Y. Tassa et al., "dm_control: Software package for physics-based simulation and RL," *arXiv:2006.12983*, 2020.

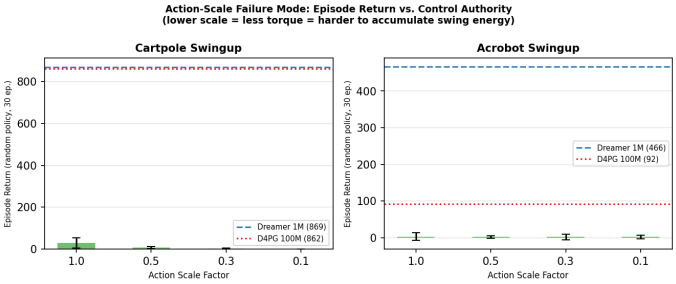

Figure 3: Action-scale failure mode (our experiment, 30 episodes each): return vs. control authority at scales 1.0, 0.5, 0.3, 0.1. Implements the energy-accumulation failure identified in M2 feedback.

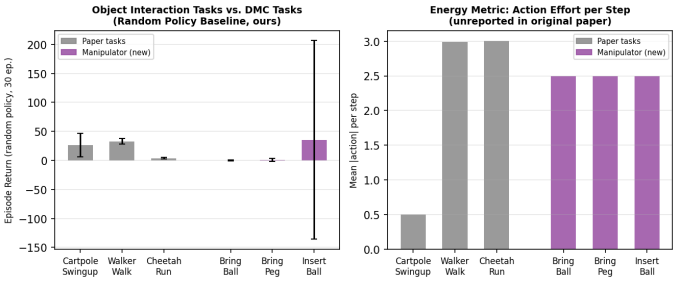

Figure 4: Left: random-policy return on paper tasks (grey) vs. manipulator tasks (purple), our experiment. Right: energy metric (mean $|a|$ per step)—unreported in paper. Implements M2 §3.2 (object interaction) and the energy metric from feedback.